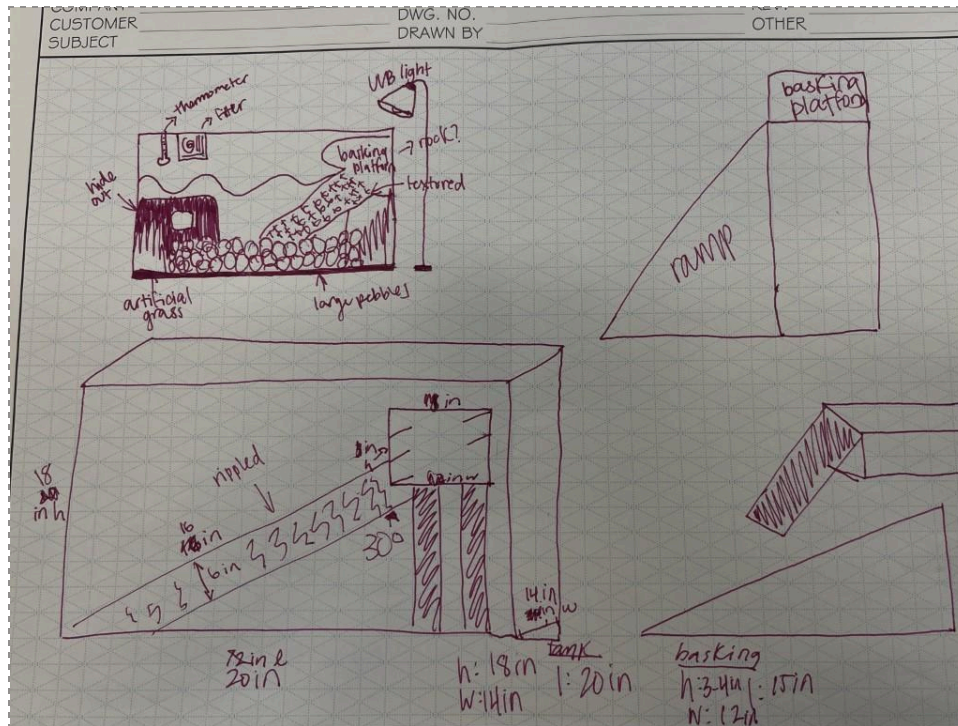


Element D



Initial blueprint with changes (2025)

Process for Generating & Comparing Ideas

- We brainstormed different ways to make an efficient tank and wrote down all ideas, and made rough sketches of each idea and focused on efficiency rather than design.
- Made changes like adjusting the ramp angle and adding texture.

Concepts we created

- Concept 1: Floating basking platform
 - Pros: adjusts to water
 - Cons: Not stable, therefore unsafe at times
- Concept 2: Enclosed UV/hot basking area above water with textured ramp (CHOSEN)
 - Pros: Safe, best heat and easy cleaning
- Concept 3: Two-level tank with land area
 - Pros: Lots of space
 - Cons: Hard to clean

Design Solution

- The design we chose met most of our high priority requirements, was safest for the turtles, ideal UV and heat levels, cleaner water due to the basking area being separated

- Gave ideal amount of space (we aimed for 12 x 18 in)

Plan of Action

1. Finalize sketch
2. Decide what material (Acrylic, cardboard, foam boards)
3. UV Bulb
4. Build cardboard prototype
5. Test what works & what doesn't
6. Rebuild if needed
7. Build Final Prototype.